

Shredded Document MEA

Instructor MEA Feedback and Assessment Package (I-MAP)

Individual Questions & Responses

Purpose of Instructor Feedback on Individual Questions:

- Enable better performance in subsequent problem solving activities (e.g. MEAs, design projects...)

High-Quality Feedback for Individual Questions:

- Identifies the pieces of the students response that are correct and incorrect
- Provides a correct response when there is only one correct response
- Provides examples of a correct response when multiple correct responses are acceptable

Context Setting: This is an exploration of some element(s) of the problem that may be unfamiliar. In this case, the ***shredding of documents and their re-assembly*** is being explored so that Problem Formulation responses and the Team Solution are more keyed in to the context of the problem.

Yes = 2	3 things learned with clear explanation of the relevance to the context of this problem <i>Potential things learned</i> and their relevance include, but are not limited to: • types of shred cuts (possible relevance: need to create an algorithm that deals with different sizes and shapes of cut) • methods for re-assembling shredded documents (possible relevance: ideas for constructing the model to solve this problem) • challenges to re-assembling shredded documents (possible relevance: ideas for things to account for when designing the model for this problem) • prosecution concerns over accuracy of re-assembly methods (possible relevance: the accuracy of the model developed for this problem needs to be high and defensible) Drop one level if there is no clear connection to the problem context
Sort of = 1	1-2 things learned; drop one level if there is no clear connection to the problem context
No = 0	No things learned 1-2 things copied directly from source (this is plagiarism) – indicate that a note has been sent to the professor

Students need to be aware of the resources for making proper citations and that ideas and information gathered in the world need to be properly cited. APA reference formats can be found at <http://owl.english.purdue.edu/owl/resource/560/01/> Most are likely to be web pages; formats include:

Author, A. A., & Author, B. B. (Date of publication). Title of document. Retrieved from <http://Web address>

Title of document. Retrieved from <http://Web address>

Yes = 2	2 citations in APA format
Sort of = 1	1 citation in APA format, 2 citations but APA format not used
No = 0	No citations, 1 citation but APA format not used

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Problem Formulation: Parts a and b ask for a big-picture view of the problem. This requires:

1. Consideration of all of the various people and organizations that care about or will be impacted by the solution to the problem and their relationship to the problem and the deliverable.
2. Consideration of issues that might arise around the implementation of the solution.

a. **List as many stakeholders as you can think of who may be impacted by the deliverable your team has been asked to create. For each stakeholder, explain the relationship between the stakeholder, the problem, and the deliverable.**

Yes = 2	Lists 5 or more of those below (or others) WITH clear explanations: FSS Software Team– the one who will use the re-assembly procedure to create a software package Federal Security Services (FSS) – consulting firm with government contracts needs a procedure to re-assemble shredded documents that is indisputable Automotive Company / Vice Presidents – under investigation; wants a fair shake Dorothy Belding, Evidence Documents Manager, FSS – supervisor for the Engineering Team, will oversee the development of the re-assembly procedure; under pressure to enable new contracts related to shredded documents Government Prosecuting Teams – want to use FSS's services and be confident in their results Public or (Potential) Victims – wants assurance that justice can be served in cases that impact their health and well-being Others Cases With Similar Needs – people and organizations that need document re-assembly and wish to be confident in the result Engineering Team – the team hired to create a document re-assembly procedure; a high-quality procedure could mean continued employment When missing explanations or explanations are inadequate, drop one level. Inadequate or vague explanations need more detail. Examples include: "X is impacted by the solution" "X will (not) benefit from the solution"
Sort of = 1	Lists 3 to 4 of above WITH clear explanations; when missing explanations drop a level
No = 0	Lists 2 or fewer

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- b. *Your solution will be implemented in the context described here and potentially in other contexts. Describe issues (minimum five) that might arise for stakeholders when your generalizable solution is implemented.*

Here your team has only been asked to consider scans of strips from single grayscale images in the development of your solution. What other issues may need to be considered when the computer-based tool that will eventually result from your team's work is implemented.

Yes = 2	Lists 4 or 5 WITH clear descriptions. Potential issues relate to issues around ... <i>Limitations of procedures applicability :</i> • Handles only single images, not multiple documents • Assumes gray-scale (no color) • Not designed to handle different image shred orientations • Not designed to handle different image shred shapes (other than rectangular shreds) • The shred images may not have clean edges <i>Pre-processing issues:</i> • Company may not be able to scan shreds in a similar fashion (data formatting issue) <i>Perceived Reliability:</i> • Can it handle more or less complex images to same degree of accuracy Unacceptable issues include • issues related to Modifiability, Re-usability, or Share-ability of the solution – these are task-level issues • issues related to the mathematics use – this is a task-level issue • restatement of the data provided • restatement of the problem itself
Sort of = 1	Lists 2 or 3
No = 0	Lists 1

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Problem Identification: Parts c to e ask for a task-level view of the problem; that is narrow the scope of the problem as viewed in Problem Formulation down to the task at hand. This requires:

1. Identification of the direct user – the person(s) who will use the deliverable.
2. Identification of the direct user's need(s), including the deliverable, its function, the criteria for success, and the constraints of the problem.
3. Probing of the problem to understand what makes creating a solution complex.

c. **Consider your list of stakeholders. Who is the direct user of the deliverable your team is being asked to create?**

Yes = 2	The FSS Software Team will actually use the procedure to develop a computer implementation.
Sort of = 1	Federal Security Services (FSS) and Dorothy Belding, Evidence Documents Manager, might need to understand the procedure. Dorothy will receive, perhaps evaluate the team's work, perhaps archive the procedure but are not likely to use the procedure. Mix of answers including FSS Software Team.
No = 0	Prosecutors, alleged criminals, and alleged victims will not use the procedure. The Applications Engineering Team (that's the student team) is not the direct user. Others with similar needs are not the immediate direct users. Mix of answers NOT including FSS Software Team.

d. **In a few sentences and in your own words, what does the direct user need? (Remember to describe the deliverable, its function, the criteria for success, and the constraints.)**

Yes = 2	A good response would be... The FSS Software Team (direct user) needs a procedure (deliverable) to re-assemble shredded documents (function) to X degree of accuracy (criteria for success) given scanned shreds image data-sets with gray-scale pixel information [255 to 0 is white to black range] (constraints). This logic must be true: deliverable & ((function & criteria for success) & constraints) Deliverable can be described using like terms for procedure (method, math model...); "a way" is too vague. Deliverable and function CANNOT be related to the re-assembled documents (the results are not the end goal) Criteria for success MUST be related to a quantitative measure of accuracy Constraints need to be described the shredded image data as being grayscale pixel information
Sort of = 1	Response must include the deliverable (a procedure, method, math model or like terms) AND the function with or without criteria for success (the quantitative measure of accuracy) OR the constraints (descriptions of shredded image data) This logic must be true: deliverable & xor(function,constraints) Constraints need to be described ONLY as grayscale pixel information but students should be encouraged to include more details (shape of data, values of pixel information)
No = 0	Response includes only the deliverable (a procedure, method, math model or like terms), cites only that results (re-assembled documents) are needed, or is not on task.

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- e. Consider the immediate problem as described and the sample data provided. Describe at least two ideas you have for why this problem might be complex to solve.

Yes = 2	2 ideas are provided with enough detail to understand. These ideas must relate to working with the data provided. Possible ideas include, but are not limited to: • Determination of "degree of" match between pixels on two different shreds • Determination of "degree of" match between two shreds • Development of method to compare all shreds to each other and sort • Determination of adequacy of accuracy of re-assembly Unacceptable ideas include restatement of the data provided or of the problem itself without some new thinking about these things
Sort of = 1	1 idea with enough detail to understand; 2 incomplete ideas
No = 0	No or 1 incomplete idea

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TEAM RESPONSE

Purpose of Instructor Feedback on Team Solutions:

Overall

- Narrow the gap between actual performance and reference level performance (indicated below). That is, encourage improvement across each dimension (below) from drafts to final response. Note that the reference level never changes from start to finish.
- Enable better performance in subsequent problem solving activities (e.g. MEAs, design projects...)

Mathematical Model

- Guide students towards identifying the complexity in the problem
- Guide students to develop models that are simple and elegant but addresses the complexity of the problem
- Guide students to look beyond simple means of comparing pixels
- Guide students to develop complete sequencing methods
- Guide students to determine the accuracy of their models and use this accuracy to improve their models

Share-ability

- Guide students towards writing a procedure that others can successfully implement
- Guide students towards presenting meaningful results that demonstrate that their model works
- Guide students towards finding a balance between providing detail and being concise

Re-usability

- Guide students to describe the task-level view of the problem and overview their solution so that others can understand when the model can be applied

Modifiability

- Guide students to engage in rational capture – articulation of decisions made to create the model
- Guide students to write evidence or context based rationales

High-Quality Feedback for Team Solutions:

- Focused on the specifics of the task, rather than on the students themselves
- Related to the students' current response (response-specific)
- Clear and simple, but elaborate enough to guide students to closing the performance gap
- Praise is NOT effective, particularly when it is mixed with the identification of problems and recommendations for improvements

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Mathematical Model

A mathematical model may be in the form of a procedure or explanation that accomplishes a task, makes a decision, or fills a need for a direct user. A high quality model fully addresses the complexity of the problem and contains no mathematical errors.

Specific to Shredded Document MEA

Looking beyond simple pixel to pixel comparisons

In a high quality model: The procedure must:

- Quantitatively determine the degree of match between pixels
Images can be complex – containing a high degree of shading variation. It might not be sufficient to consider just the degree of match between a left strip, right edge pixel and a right strip, left edge pixel. It may require looking at the degree of match to neighbors. Consider the shredded document strip data below. Just looking at the difference between the 64-255 match might lead one to believe that the match is not very strong, yet the neighboring pixels are of a similar value to the 64.

Left Strip, Right Edge -1	Left Strip, Right Edge	Right Strip, Left Edge
254	255	255
255	255	255
255	255	255
64	255	62
63	63	64
50	64	255
64	62	64
67	255	59
255	255	255
255	255	255

- Quantitatively determine the degree of match between shreds. If there are many shreds, will there be enough distinction between shred matches?
- Compare one shred to every other shred (left and right side)
- Sort shreds according to degree of match between shreds
- Provide a mechanism for breaking ties, if ties are likely to occur (given the procedure)
- Provide a quantitative assessment of the model

Solutions to Image Provided:

Draft 1:

image0 – order should be C,F,E,B,G,D,H,A or G,D,H,A,C,F,E,B (the cut between shreds is between the two letters, so the order could be G6 or 6G).

Draft 2:

image 0 – as above

image1 – B, J, I, G, D, H, C, F, E, A, K

Team Final Response: image0 (as above) and image1 (as above) and two images from the individual data generation step

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LEVEL 0 –

- There is no quantitative procedure to re-assemble the shredded document image (e.g. procedures involving visual inspection only are not quantitative)

LEVEL 1 -

- Only *simple equality* of strip pixels is used to determine pixel matches

LEVEL 2 –

- Only *simple difference* between strip pixels is used to determine pixel matches
- Mathematical detail may be lacking or missing.
- Mathematical errors might be present.
- If the solution demonstrates lack of understanding of the context of the problem, this is the highest level achievable.

LEVEL 3 –

- Degree of difference between strip pixels is used to determine pixel matches
- Applying a method to quantify the degree of shred-to-shred match (missing – drop a level)
- Applying a method to compute all shred-to-shred matches (left-rights and right-lefts) (missing – drop a level)
- Applying a method to order the shreds to re-assemble the document (missing – drop a level)
- The procedure provides a viable strategy to break a tie – *IF the method could potentially have ties* (missing – drop a level)
- The procedure includes a method to quantitatively assess the accuracy of the model (missing – drop a level)
- Some mathematical detail may be lacking or missing.
- Mathematical errors might be present.
- If there is an indication that the team does not understand one or more statistical measures being used, drop to the next level

LEVEL 4 –

- Accounting for pixel neighbors
- Applying a method to quantify the degree of shred-to-shred match (missing – drop a level)
- Applying a method to compute all shred-to-shred matches (left-rights and right-lefts) (missing – drop a level)
- Applying a method to order the shreds to re-assemble the document (missing – drop a level)
- The procedure provides a viable strategy to break a tie – *IF the method could potentially have ties* (missing – drop a level)
- The procedure includes a method to quantitatively assess the accuracy of the model (missing – drop a level)
- Handles ties
- The procedure provides a viable strategy for how to break tie.
- Mathematical detail should be clear from start to finish.
- Mathematical errors should be eliminated.

Accounting for Data Types

It must be determined whether the mathematical model *takes into account all types of data provided* to generate results. If any shredded document images are not used in the mathematical model, a reasonable justification must be provided.

LEVEL 4 – All shredded document images are used OR clear justification for discarding images are provided

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LEVEL 3 – All shredded document images are NOT used and no justifications are provided. Justifications might be provided for things like:

- Image is all white – these might be image edges and don't contribute to overall image re-assembly

Generalizability

Generally, one would not produce a mathematical model to solve a problem for a single situation. A mathematical model is produced when a situation will arise repeatedly, with different data sets. Therefore, the model needs to be able to work for the data set provided and a variety of other data sets. That is, a useful mathematical model is adaptable to similar, but slightly different, situations. For example, a novel data set may emerge that wasn't accounted for in the original model, and thus the user would need to revise the model to accommodate the new situation.

A mathematical model that is generalizable is share-able, re-usable, and modifiable. Thus, one should strive for clarity, efficiency and simplicity in mathematical models; as such models are the ones that are more readily modified for new situations. Although the student team has been "hired" as the consultant team to construct a mathematical model, direct user needs and wants to understand what the model accomplishes, what trade-offs were involved in creating the model, and how the model works.

Re-Usability

Re-usability means that the procedure can be used by the direct user in new but similar situations.

A *re-usable* procedure:

- Identifies who the direct user is and what the direct user needs in terms of the deliverable, its function, criteria for success, and constraints
- Provides an overarching description of the procedure
- Clarifies assumptions and limitations concerning the use of procedure. These include assumptions about the situation and the types of data to which the procedure can be applied. ***There ARE limitations built in to the problem description; these must be a stated***

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Re-Usability Item	Shredded Document	Yes (2 pts)	Sort Of (1 pt)	No (0 pt)
Identification of direct user	FSS's Software Team			
Deliverable	Procedure			
Criteria for Success	To re-assemble shredded documents to an appropriate accuracy goal for overall document re-assembly (a quantitative measure is needed here. For example, a goal might be on the order of 80-90% match to actual)	re-assembly AND accuracy goal	re-assembly OR accuracy goal	
Constraints	Given scanned image data (pixel information) of shreds. The pixel information must be gray-scale (255-white to 0-black)		missing pixel value info	
Overarching Description	Should provide an overview of how the image sequencing is determined and the level of accuracy of procedure.	procedure AND accuracy	procedure OR accuracy	
Assumptions and limitations concerning the use of procedure	Limitations: - for full, clean shreds - from one image (one page) - grayscale data - strips in same orientation Other limitations will depend on the details of the procedure.	3 or more limitations	1-2 limitations	

LEVEL 4: rubric score of 9-12

LEVEL 3: rubric score of 6-8

LEVEL 2: rubric score <= 5

Modifiability

Modifiability means that the procedure can be modified easily by the direct user for use in different situations.

A modifiable procedure:

- Contains acceptable rationales for critical steps in the procedure and
- Clearly states assumptions associated with individual procedural steps.

Given this type of information, the direct user will be able to modify (change) the model for new situations.

Critical steps that need justification / rationale:

- Pixel matching method
- Shred matching method
- Sorting method
- Degree of success at re-assembly of a document
- Hard-coded values imbedded in procedural steps require explicit explanation of where the values come from.

Rationales are tied to the mathematical model. So students need to be reminded that when their model changes, they need to revise, delete, and add rationales to make them appropriate for their model.

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Share-ability

Share-ability means that the direct user can apply the procedure and replicate results. If the mathematical model is not developed in enough detail to clearly demonstrate that it works on the data provided, it cannot be considered shareable.

Results

NOTE: On Team Final Response, the team should include 2 team created data image sets in the results section.

LEVEL 4 achievement requires that the mathematical model be applied to the data provided to generate results in the form requested. Quantitative results are to be provided.

Results of applying the procedure MUST be included in the memo.

LEVEL 1 – No results or order of shreds with no quantitative results or results do not seem to be those for the data set indicated. Ensure that the student teams are presenting results for the specified data set. Multiple data sets may have been made available to the students and the analysis of only the latest may have been requested in the current memo.

LEVEL 2 – Some quantitative results, but inadequate to determine the quality of the procedure. Significant figures are a problem.

LEVEL 4 –

Draft 2: For image0 and image1: correct order of shreds, order of shreds generated by algorithm, degree of match between shreds (or other intermediate quantitative measure), and quantitative measure of accuracy are provided. (missing any drop to level 2)

Team Final: For image0, image1, and 2 team created data sets: correct order of shreds, order of shreds generated by algorithm, degree of match between shreds (or other intermediate quantitative measure), and quantitative measure of accuracy are provided. (missing any drop to level 2)

Apply and Replicate Results

NOTE: On Team Final Response, the team should include an explanation of how those image sets are different from the ones provided and therefore how they further test the team's solution in ways the provided two data sets do not.

A high quality product (i.e., model communicated to the direct user) will clearly, efficiently and completely articulate the steps of the procedure. A high quality product may also illustrate how the model is used on a given set of data. The description will be clear and easy to follow; it must enable the results of the test case to be reproduced. At a minimum, the results from applying the procedure to the data provided must be presented in the form requested.

The direct user requires a relatively easy-to-read-and-use procedure. If this has not been delivered, the solution is at most LEVEL 3 work.

If you, as a representative of the direct user, cannot generate results, the solution is not LEVEL 3 work.

Team Final: Drop one level if missing explanation of how the 2 team image sets are different from the ones provided and therefore how they further test the team's solution in ways the provided two data sets do not.

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Extraneous Information

The mathematical model should be free of distracting and unnecessary text. This might include (1) outline formatting, (2) indications of software tools (e.g. MATLAB or Excel or, more generally, spreadsheets) necessary to carry out computations, (3) explicit instructions to carry out common computations, (4) discussions of issues outside the scope of the problem, and (4) general rambling.

LEVEL 3 – If any of the following are present::

- *Outline formatting*
- *Mentions of computer tools*
- *Descriptions of how to compute standard measures (e.g. mean, standard deviation)*
- *Excessive discussions centered on big picture issues are not necessary*

Background Information

This document contains MEA2 information for TAs. This MEA is about **Shredded Documents**.

As before, TAs need to provide their own solution to Content Setting Questions, Individual Questions, and a solution to Draft 2.

Document Information:

This document contains the following information:

- Page 1: Background Information and Document Information
- Page 1: Directions for TAs and Due Dates
- Page 2-3: Context and Individual Questions
- Page 4-5: Draft 1 Memo
- Page 6: Draft 2 Memo
- You will be provided with a zip file **MEA2_ShreddedDoc_DataFiles.zip** on Blackboard in Files for TAs > MEAs. You will need Draft 2 data for your solution.

Directions for TAs:

1. Read the memos on pages 5 thru 7.
2. For this MEA, you will go to “MEA 2 TA Training” in the “Spring 2012 – ENGR 132 TA Training” section which is available on-line at <http://www.mealearning.com>.
3. Answer the “Content and Individual Questions” on-line at <http://www.mealearning.com>.
4. Provide your own solution to Draft 2 Memorandum on-line at <http://www.mealearning.com>.

Due Dates:

1. Content and Individual Questions and your solution to Draft 2 Memorandum are due on-line at <http://www.mealearning.com> by Tues, Feb 14th at 5pm.
2. On Feb 14th at 5pm, the MEA2 rubric will be emailed to TAs. All TAs must read the rubric and provide an assessment of your solution by the start of training on Feb 21st or Feb 23rd. In addition before training, TAs must also complete some practice grading for Context and Individual Questions. Details will be provided in a separate email.
3. MEA2 training will be held as follows:
 - o Tuesday, Feb 21st: 6-8:30pm, ARMS B098
 - o Thurs, Feb 23rd: 6-8:30pm, ARMS B098
 - o **All TAs must attend one of these above training sessions (not both). This is mandatory for all TAs.**
 - o During training, the rest of the MEA schedule will be explained.

Context and Individual Questions

1. Individually read the two memos from Dorothy Belding.
2. Access the MEA portal and answer the Context Setting and Individual Questions at:
<https://www.mealearning.com>
3. Document shredders are a standard item in a busy office. What kinds of shredding can one expect from such devices? What challenges do these shreds present for re-assembly? What are issues around using re-assembled documents as evidence of a crime?

Use and document (with proper citations) at least **two** external resources to learn **three** things about document shredding and re-assembly that are relevant to this problem context.

In the Context Setting box on the MEA 1 interface, list, ***in your own words***, the **three** things you learned and explain how each is relevant to this problem context. In the same box, below the three things learned, list your citations using APA format. (For help with APA reference formats, see <http://owl.english.purdue.edu/owl/resource/560/01/>)

4. Individually, answer the following questions.

Problem Formulation: Parts a and b ask you to take a big-picture view of the problem.

- a. List as many stakeholders as you can think of who may be impacted by the deliverable your team has been asked to create. For each stakeholder, explain the relationship between the stakeholder, the problem, and the deliverable.
- b. Your solution will be implemented in the context described here and potentially in other contexts. Describe issues (minimum five) that might arise for stakeholders when your generalizable solution is implemented.

Here your team has only been asked to consider scans of strips from single grayscale images in the development of your solution. What other issues may need to be considered by FSS as it implements the computer program that will eventually result from your team's work.

Problem Identification: Parts c to e ask you to take a task-level view of the problem.

- c. Consider your list of stakeholders. Who is the direct user of the deliverable your team is being asked to create?
- d. In a few sentences and in your own words, what does the direct user need? (Remember to describe the deliverable, its function, the criteria for success, and the constraints.)
- e. Consider the immediate problem as described and the sample data provided. Describe at least two ideas you have for why this problem might be complex to solve.

You may wish to consider the small problem below as you think about the complexity of the larger problem.

Given the following values for the edges of two strips, develop a method for quantitatively rating how well these two pieces align. Be sure to indicate how to interpret your rating.

Left Strip, Right Edge	Right Strip, Left Edge
255	255
255	255
255	255
255	62
63	64
64	255
62	64
255	59
255	255
255	255

Draft 1 - Memorandum

To: Engineering Team
From: Dorothy Belding, Evidence Documents Manager, Federal Security Services
Re: Shredded Document Recovery Process

Federal Security Services (FSS) is a freelance security consulting firm with numerous governmental contracts. Due to the recent increase in illegal business practices, state and federal government prosecuting teams have been contacting us to aid in their investigations. We need to develop a structured method for the recovery of shredded documents. The high profile nature of many of these cases has resulted in more intense scrutiny on how evidence of this nature is obtained and recovered. Currently, few formally documented processes exist for the recovery of shredded documents, causing key documents to be excluded as evidence due to the potential for tampering. FSS would like to develop a computer-based tool to handle document recovery.

The picture shown in Figure 1 is a digitized, gray scale image. This image happens to be a view seen in an optical microscope, but similar images are currently being created of scanned document strips. An image is stored as a data file which contains an array of numbers where each number refers to the gray scale value for each pixel of the image. A sub-sample of the complete Figure 1 file is shown in Figure 2 and Table 1. This sub-sample shows the gray scale values for the square superimposed on Figure 1 and exploded by itself in Figure 2. Gray scale values range from 0 for black to 255 for white.

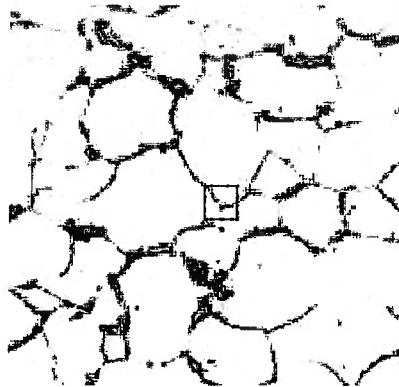


Figure 1. Micrograph of crystals with sub-sample shown.

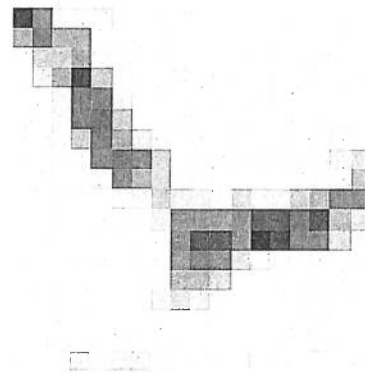


Figure 2. Sub-sample of micrograph shown in Figure 1.

Table 1. Digital file for sub-sample from micrograph of crystals shown in Figure 2.

128	160	221	221	221	241	241	234	241	255	241	241	241	241	255	255	234	234
178	153	178	178	227	234	234	234	255	241	234	234	234	241	241	241	241	241
234	192	192	178	221	227	234	234	255	241	234	241	234	241	234	234	255	241
241	204	178	134	192	221	234	227	234	234	234	241	234	255	234	234	241	234
234	234	215	153	160	215	234	227	227	234	234	241	234	255	234	234	234	227
227	241	221	153	150	192	234	234	234	234	234	234	234	241	234	241	234	227
234	241	221	192	153	178	204	221	234	234	227	234	234	241	227	241	221	221
241	234	227	227	160	153	153	192	227	227	221	234	234	227	221	234	204	192
241	234	234	227	215	160	178	192	234	221	227	227	227	227	227	221	215	178
241	234	234	234	227	215	215	192	204	204	215	192	204	204	192	178	153	150
241	241	241	234	227	241	234	204	160	160	160	153	134	134	150	134	178	192
241	241	241	234	227	234	227	221	153	134	134	153	119	128	153	153	192	221
241	241	241	234	234	227	221	221	160	150	153	192	204	215	234	221	227	234
241	241	241	234	241	234	227	215	178	178	192	221	227	227	234	221	227	234
241	241	234	234	234	234	234	204	192	204	227	227	241	234	234	234	241	241
241	234	234	234	234	227	234	221	227	221	234	227	227	234	234	241	227	234
234	241	227	241	221	215	221	227	227	227	234	241	234	234	255	227	234	241
234	234	241	204	215	204	204	215	227	234	227	227	255	227	241	255	241	221

In a true black and white document, such as a text-only document, the gray scale values present will be limited to 0 (black) and 255 (white). A perfectly black and white image will contain ONLY 0's and 255's.

The gray-scale value of each pixel is accessible information that can be used to make decisions about whether or not document strips fit together.

Last week, a government raid on a major automotive company resulted in the seizing of 8 bags of shredded documents. Though the majority of the documents are most likely legitimate confidential business memos, there is evidence to suggest that some documents were shredded to cover up indications that a high ranking vice-president was aware of poor product performance which resulted in a number of deaths.

I would like your engineering team to develop the algorithm that the programming team will use to reassemble the shredded documents. The FSS programming team will take your written description and translate it to software code. Because the FSS programming team is still evaluating several potential programming languages in which to implement the project, you should avoid using any specific programming language to describe your procedure.

Dorothy Belding

Draft 2 - Memorandum

To: Engineering Team
From: Dorothy Belding, Evidence Documents Manager, Federal Security Services
Re: Shredded Document Recovery Process (Update)

To help your team in developing an algorithm to reassemble shredded documents, I have asked the FSS programming team to create a test case for you to manipulate and test your procedure. Here is the information they have sent to me:

Because the complete set of pixel information for an entire document is so large, the test case that we have provided is a couple of characters from a set of scanned shreds of grayscale document. The scanned shreds are provided in two formats for your convenience: (1) 8 individual shreds saved as eight .dat files called **im0shredA.dat** to **im0shredH.dat**, and (2) an Excel file called **image0_f11.xlsx** into which each .dat file was loaded into a separate tab. In the Excel file, conditional settings are used to gray scale shade the cells containing data according to the pixel value. This allows you to have visual cues about the test data. If you set the View>Zoom to 10%, you can see a decently sharp image of the scan. As an alternative, the data in the .dat files can viewed in MATLAB using the following code:

```
>> shredA = load('im0shredA.dat') % loads shredA data
>> shredA_image_matrix(:,:,1)=shredA % these three lines create
>> shredA_image_matrix(:,:,2)=shredA % a viewable image matrix
>> shredA_image_matrix(:,:,3)=shredA
>> imwrite(shredA_image_matrix) % creates an image of shredA in a figure window
```

Utilizing this information, please develop an algorithm to reassemble gray scale documents. The FSS programming team has specifically requested that you pay special attention to describing how you will decide whether two (2) strips should be joined together. Your final procedure should describe the entire process for reconstructing a document from a set of scanned strips. A quantitative assessment of the model should also be provided.

In a memo to my attention, please include your team's procedure, the order of the scanned strips (A to H) generated by applying your procedure to the test data, and a quantitative assessment of your model. Please be sure to include your team's reasoning for the each step, heuristic (i.e. rule), or consideration in your team's procedure.

Please send your complete memo to me by next Tuesday.

Dorothy Belding